

Safe C++ @ O'Reilly

Klaus Iglberger
February, 3rd, 2026

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1. Introduction

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Introduction of the Trainer

C++ Trainer/Consultant since 2016

Author of “C++ Software Design”

(Co-)Organizer of the Munich C++ user group

(Co-)Chair of the Software Design track at CppCon

Regular presenter at C++ conferences



Klaus Iglberger

Scope of this Training

In this training I will address ...

- ... the perceived safety problem of C++
- ... a discussion of possible solutions for those safety problems
- ... the philosophy of “Modern C++”

Content of this Training

1. Introduction
2. Safe C++
3. Summary/Literature

Schedule

Tuesday, February, 3rd, 2026 (EDT time zone)

10:00am - 10:10am	<i>Introduction</i>
10:10am - 10:30am	<i>Motivation: The Many Safety Problems of C++</i>
10:30am - 11:00am	<i>Resolving Undefined Behavior with constexpr</i>
11:00am - 11:10am	<i>Break</i>
11:10am - 11:40pm	<i>Resolving Out-Of-Bounds Problems with Ranges</i>
11:40am - 12:00pm	<i>Resolving Type Safety with Strong Types (I)</i>
12:00pm - 12:10pm	<i>Break</i>
12:10pm - 12:30pm	<i>Resolving Type Safety with Strong Types (II)</i>
12:30pm - 1:00pm	<i>Resolving Memory Safety with Value Semantics (I)</i>
1:00pm - 1:10pm	<i>Break</i>
1:10pm - 1:30pm	<i>Resolving Memory Safety with Value Semantics (II)</i>
1:30pm - 1:55pm	<i>Dealing with the Sharp Edges</i>
1:55pm - 2:00pm	<i>Summary</i>

Miscellaneous

I assume you have some experience with C++. If something remains unclear or somewhat vague, **please ask!**

Miscellaneous

Also remember: **You are in control!**

Guidelines

(Core) Guideline: ...

The green boxes represent ...

- ... the most important take-aways;
- ... common idioms and best practice;
- ... markers in the slides.

Programming Tasks

Task (Subchapter/Name): ...

The yellow boxes represent ...

- ... hands-on programming tasks;
- ... links to the provided source code examples.

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